

Tutorial 2

PS1-6: m.v. Gaussian

CS5487 Lecture Notes (2024A)
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$x \in \mathbb{R}^d$, mean $\mu \in \mathbb{R}^d$, $\Sigma \in \mathbb{S}_{++}^d$ covariance

$$p(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} e^{-\frac{1}{2} \|x - \mu\|_{\Sigma}^2} = N(x | \mu, \Sigma)$$

Mahal. distance: $\|x - \mu\|_{\Sigma}^2 = (x - \mu)^T \Sigma^{-1} (x - \mu)$

(a) let Σ be a diagonal matrix

$$\Sigma = \begin{bmatrix} \sigma_1^2 & & 0 \\ & \ddots & \\ 0 & & \sigma_d^2 \end{bmatrix} \Rightarrow |\Sigma| = \prod_{i=1}^d \sigma_i^2 \quad (\text{product of diagonal elements})$$

$$\Sigma^{-1} = \begin{bmatrix} \sigma_1^{-2} & & 0 \\ & \ddots & \\ 0 & & \sigma_d^{-2} \end{bmatrix} \Rightarrow \|x - \mu\|_{\Sigma}^2 = (x - \mu)^T \begin{bmatrix} \sigma_1^{-2} & & 0 \\ & \ddots & \\ 0 & & \sigma_d^{-2} \end{bmatrix} (x - \mu)$$

$$= (x - \mu)^T \begin{bmatrix} \frac{1}{\sigma_1^2} (x_1 - \mu_1) \\ \vdots \\ \frac{1}{\sigma_d^2} (x_d - \mu_d) \end{bmatrix}$$

$$= \sum_{i=1}^d \frac{1}{\sigma_i^2} (x_i - \mu_i)^2$$

Plug-in:

$$p(x) = \frac{1}{(2\pi)^{d/2} \prod_{i=1}^d \sigma_i} e^{-\frac{1}{2} \sum_{i=1}^d \frac{1}{\sigma_i^2} (x_i - \mu_i)^2} = \prod_{i=1}^d \left(\frac{1}{(2\pi)^{1/2} \sigma_i} e^{-\frac{1}{2\sigma_i^2} (x_i - \mu_i)^2} \right)$$

$\underbrace{\prod_{i=1}^d e^{-\frac{1}{2} \frac{1}{\sigma_i^2} (x_i - \mu_i)^2}}_{\substack{\text{univariate Gaussian} \\ \text{for } x_i}}$

$$p(x) = \prod_{i=1}^d N(x_i | \mu_i, \sigma_i^2)$$

• Since the joint $p(x)$ is the product of marginals $p(x_i)$, then $x_i \perp x_j$, $i \neq j$. Also, $\sigma_{ij}^2 = 0$, covariance b/w x_i & x_j is 0.

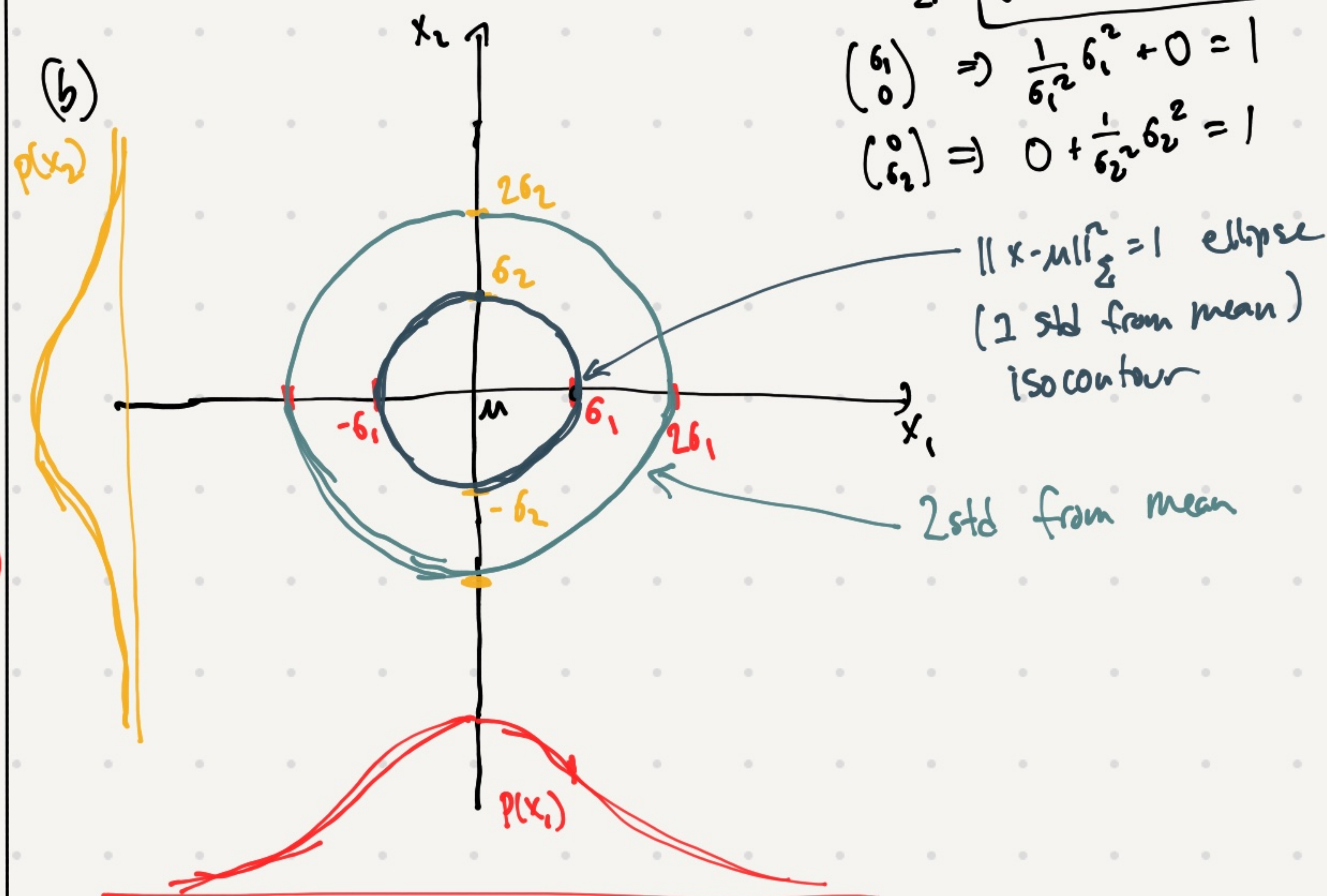
$$\|x - \mu\|_{\Sigma}^2 = \sum_i \frac{1}{\sigma_i^2} (x_i - \mu_i)^2 = 1$$

$$\begin{pmatrix} \sigma_1 \\ 0 \end{pmatrix} \Rightarrow \frac{1}{\sigma_1^2} \sigma_1^2 + 0 = 1$$

$$\begin{pmatrix} 0 \\ \sigma_2 \end{pmatrix} \Rightarrow 0 + \frac{1}{\sigma_2^2} \sigma_2^2 = 1$$

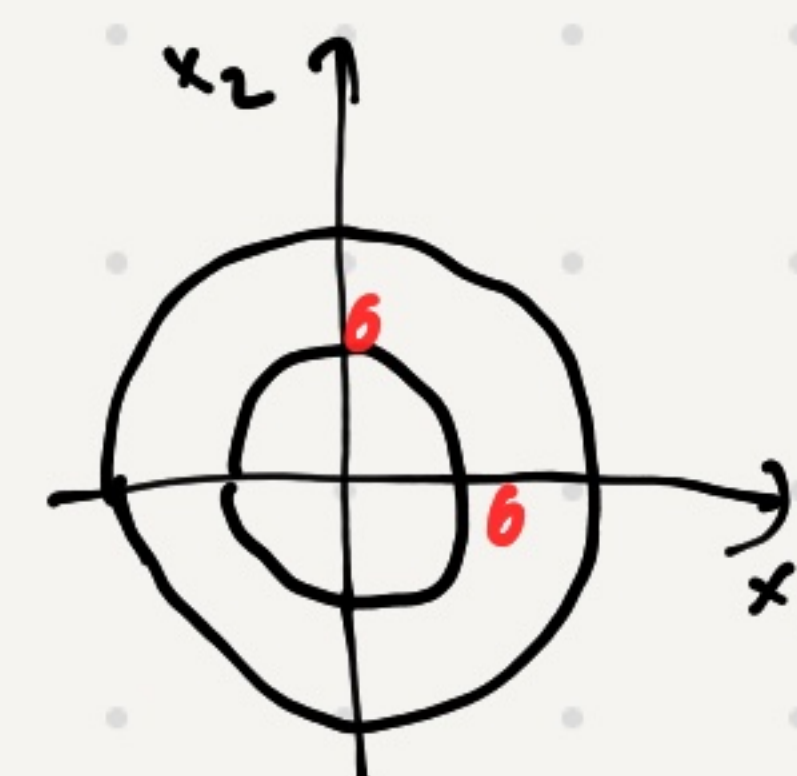
$\|x - \mu\|_{\Sigma}^2 = 1$ ellipse (1 std from mean) isocontour

2std from mean



Special case: if $\Sigma = \sigma^2 I$ (isotropic covariance, iid covariance)

$\Rightarrow$ circular Gaussian



(d) General Case Σ

$$\Sigma v = \lambda v \quad \leftarrow \text{special vector where } \Sigma v, \text{ I get back a scaled version of } v.$$

$\uparrow$ eigenvector $\uparrow$ eigenvalue

$$(\Sigma - \lambda I) v = 0$$

For $\Sigma \in S_{++}^{d \times d} \Rightarrow d$ eigenvector/value pairs (λ_i, v_i)

$$\Rightarrow \Sigma v_i = \lambda_i v_i \quad i=1 \dots d$$

$$\Rightarrow \Sigma \underbrace{\begin{bmatrix} v_1 & \dots & v_d \end{bmatrix}}_{\substack{V \\ \text{matrix of eigenvectors}}} = \underbrace{\begin{bmatrix} \lambda_1 v_1 & \dots & \lambda_d v_d \end{bmatrix}}_V = \underbrace{\begin{bmatrix} v_1 & \dots & v_d \end{bmatrix}}_V \underbrace{\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_d \end{bmatrix}}_{\substack{\Delta \\ \text{diag mtr of eigenvalues.}}}$$

$$\Rightarrow \Sigma \underbrace{V}_{V^{-1}} = \underbrace{V \Delta}_{V^{-1}} \quad \leftarrow \text{post-multiply by } V^{-1}$$

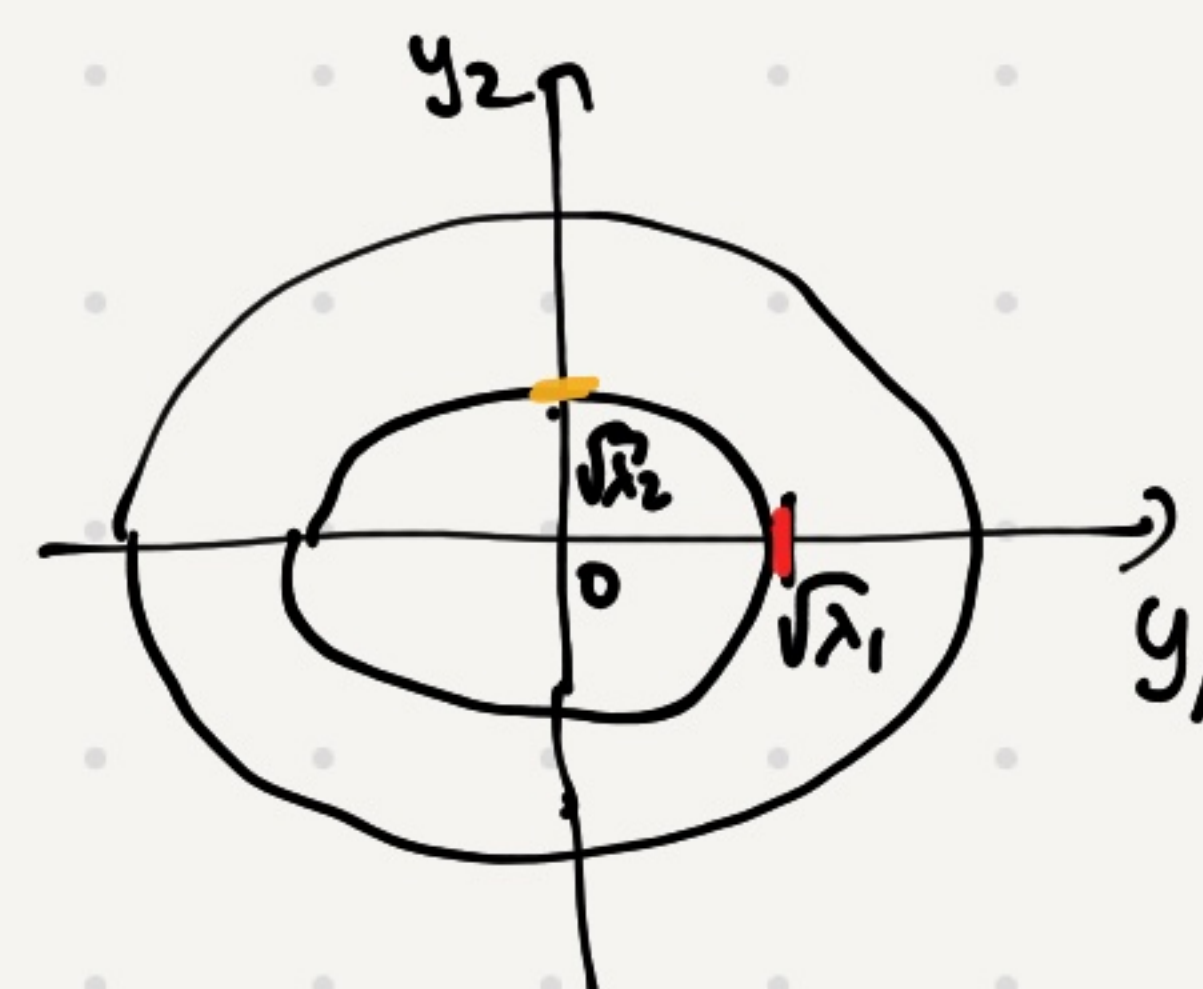
$$\Sigma = V \Delta V^{-1} \quad \leftarrow \begin{matrix} V^T V = I \text{ (orthogonal)} \\ \Rightarrow V^{-1} = V^T \end{matrix}$$

$$\Rightarrow \boxed{\Sigma = V \Delta V^T} \quad \leftarrow \text{eigen decomposition}$$

$$\Sigma^{-1} = (V \Delta V^T)^{-1} = (V^T)^{-1} \Delta^{-1} V^{-1}$$

$$\Rightarrow \boxed{\Sigma^{-1} = V \Delta^{-1} V^T} \quad \leftarrow \begin{matrix} (AB)^{-1} = B^{-1} A^{-1} \\ V^{-1} = V^T, (V^T)^{-1} = (V^{-1})^{-1} = V \end{matrix}$$

$$\begin{aligned} \|x - \mu\|_{\Sigma}^2 &= (x - \mu)^T \Sigma^{-1} (x - \mu) \\ &= \underbrace{(x - \mu)^T V}_{y^T} \underbrace{\Delta^{-1} V^T (x - \mu)}_{y = V^T (x - \mu)} \\ &= y^T \underbrace{\Delta^{-1}}_{\substack{\text{diagonal matrix}}} y \end{aligned}$$



rewrite: $y = V^T (x - \mu)$

$$V y = x - \mu$$

$$\Rightarrow x = \underbrace{V y}_{\substack{\text{rotation}}} + \underbrace{\mu}_{\substack{\text{translation}}}$$

